Addis Ababa Science & Technology University
Department of Physics

General Physics (phys1001)

Chapter - 4

Heat & Thermodynamics

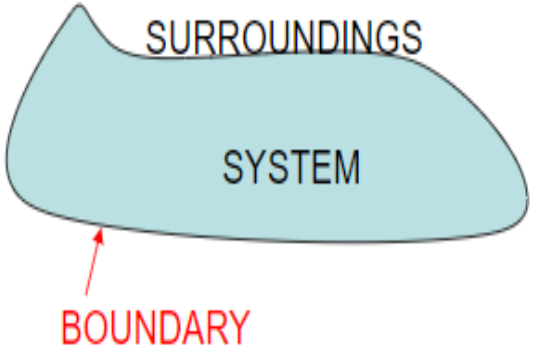
Heat & Thermodynamics

Thermodynamics :

- Is a branch of physics, which deals with the relationship among heat, work, temperature, energy and entropy.
- In broad terms, it deals with the transfer of energy from one system to another and from one form to another.

Thermodynamic system :

- It is quantity of matter or a region in space under study, where thermodynamic processes occur.
- It serves as the central focus for analyzing energy and heat transfer.
- Open System: Allows both energy and matter to flow in and out
- Closed System: Allows energy but not matter to cross its boundaries.
- Isolated System: Does not allow energy or matter to cross its boundaries.



The surrounding :

- Everything outside the system that can interact with it, these interactions can be exchanges of heat or work.
- It is defined by its temperature, volume, pressure, and chemical composition.

Note: A system is at equilibrium, when each of these variables has the same value at all points.

Heat (Q):

- It is the transfer of energy across the boundary of a system due to a temperature difference between the system and its surroundings
- It is the flow of energy from one object to another.
- Heat flows from higher temperature to lower temperature.
- Heat transfer obeys the law of conservation of energy (if no heat is lost to the surroundings)

$$Q_{\text{lost by hotter object}} = Q_{\text{gained by cooler object}}$$

$$Q = Cm\Delta T$$

Work (W):

- It is a **non-spontaneous** transfer of energy into or out of a system due to force acting through a displacement.
- It takes many forms, moving a piston, or stirring, or running an electrical current through a resistance.

Note: Heat and work are two possible ways of transferring energy from one system to another.

Temperature(T):

- It measures the average molecular kinetic energy of a substance.
- It indicates the hotness or coldness of a substance.
- It determines the direction of heat flow (i.e., Heat flows from higher temperature to lower temperature)

Thermal equilibrium:

- It is a state in which two or more systems reach at the same temperature scale.
- At thermal equilibrium, there is no net heat transfer between the system.

$if \quad T_A = T_B$, then A and B are in thermal equilibrium

Internal Energy :

- It is a total energy contained within a system , which is associated with microscopic components of atoms and molecule.
- Internal energy includes: kinetic energy of random translational, rotational, and vibrational motion of molecules, potential energy within molecules, and potential energy between molecules.

Thermal Energy and Bond Energy

- Thermal energy : can be interpreted as that part of the internal energy associated with random motion of molecules , which related to temperature and kinetic energy at the microscopic level .
- Bond energy : is the intermolecular potential energy in an atoms.

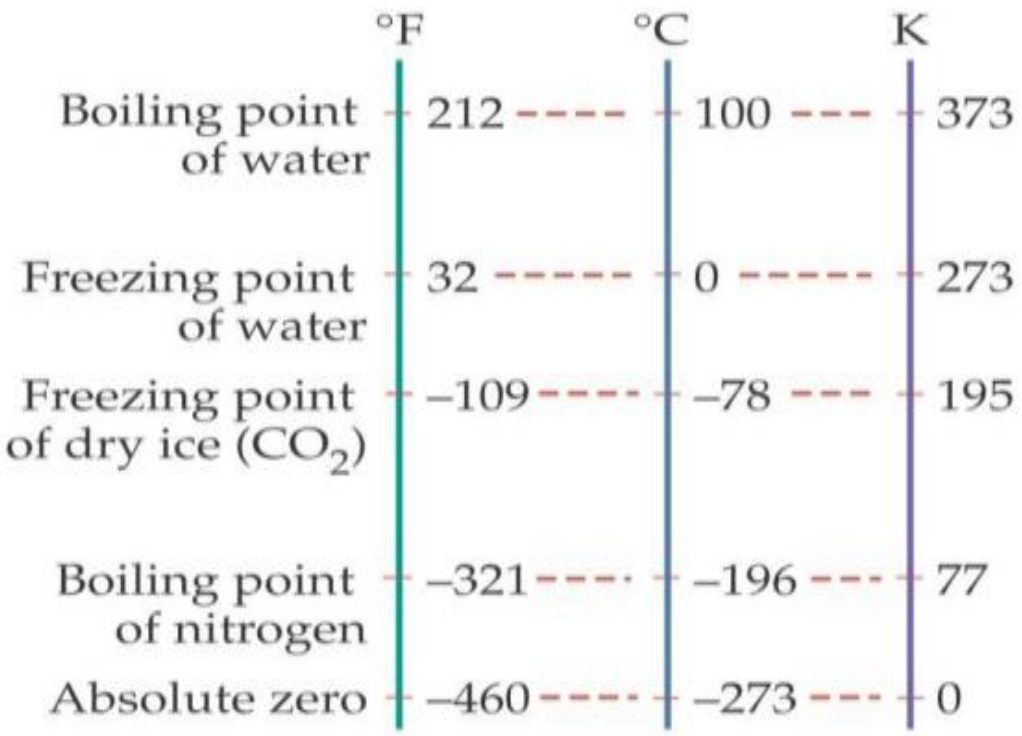
Internal energy = Thermal energy + Bond energy

Temperature Scales

- Thermometer: measures temperature according to well-defined scales of measurement
- There are three most common temperature scales : Fahrenheit, Celsius, and Kelvin.
- at STP:
  - the boiling point of water = $100^o c = 212^o F = 373^o K$
  - the freezing point of water = $0^o c = 32^o c = 273^o c$
- $Note: \Delta T_C = \Delta T_F = \Delta T_K$

Relationships between Temperature Scales

To convert from...	Use this equation...
Celsius to Fahrenheit	$T_F = \frac{9}{5}T_C + 32$
Fahrenheit to Celsius	$T_C = \frac{5}{9}(T_F - 32)$
Celsius to Kelvin	$T_K = T_C + 273.15$
Kelvin to Celsius	$T_C = T_K - 273.15$
Fahrenheit to Kelvin	$T_K = \frac{5}{9}(T_F - 32) + 273.15$
Kelvin to Fahrenheit	$T_F = \frac{9}{5}(T_K - 273.15) + 32$



Thermal Expansion

- It is the change in size or volume of a given system due to its temperature changes

1. Linear Thermal Expansion:

- The increase in length Δl of a solid is proportional to its initial length l_o and the change in its temperature ΔT .

$$\Delta l \sim l_o \Delta T$$

$$\Delta l = \propto l_o \Delta T \quad \Rightarrow l_f - l_o = \propto l_o \Delta T \quad \Rightarrow l_f = l_o + \propto l_o \Delta T = l_o(1 + \propto \Delta T)$$

$$\alpha = \frac{\Delta l}{l_o \Delta T},$$ is called coefficient of linear expansion and its SI unit $= \frac{1}{K} = K^{-1} = (C^0)^{-1}$

Where., $l_o =$ the initial length, $T_o =$ initial temperature, $l_f = l =$ final length final length

$\Delta T = T - T_0$

Example:

1. A segment of steel railroad track has a length of 30.0 m when the temperature is 0.0^oC. What is its length when the temperature is 40.0^oC? use : $\alpha = 11 \times 10^{-6} (^\circ C)^{-1}$

2. Areal Expansion

The change in area ΔA of a solid is proportional to its initial area A_0 and thechange in its temperatureΔT. That is,

$\Delta A = \beta A_0 \Delta T \implies A = A_0(1 + \beta \Delta T)$, where $\beta = 2\alpha$ is the coefficient of areal expansion.

3. Volume Expansion

The change in volumeΔV of a solid is proportional to its initial volume V_0 and the change in its temperature ΔT. That is:

$\Delta V = \gamma V_0 \Delta T \implies V = V_0(1 + \gamma \Delta T)$, where $\gamma = 3\alpha$ is the coefficient of volume expansion.

Example

1. If the temperature of a cube made of copper with initial volume $2m^3$ changes from 15^oc to 25^oc . Calculate the change in volume of the cube (use: $\alpha = 17x10^{-6}\ k^{-1}$)

Specific Heat and Latent Heat

1. Specific Heats(Specific Heat Capacity) [c → small letter]:

- It is the amount of the heat required to raise the temperature of a 1kg mass of a substance by 1^oC.

- The quantity of heat, **Q**, required to change the temperature of a body of mass **m** by ΔT is proportional to both the mass, m and the change in temperature, ΔT. And it is given by:

$$Q \sim m\Delta T \quad \Rightarrow \quad Q = mc\Delta T$$

Where, $\boldsymbol{c} = \frac{\boldsymbol{Q}}{\boldsymbol{m}\,\Delta\boldsymbol{T}}$ is specific heat capacity,

$$\text{Its SI unit} = \frac{\text{J}}{\text{kg. K}} \text{ or } \frac{\text{J}}{\text{Kg. } o_C}$$

2. Heat capacity(C) [C → Capital letter]:

- It is the amount of heat energy required to raise the temperature of the whole mass of a substance by 1^oC

$$C = \frac{\boldsymbol{Q}}{\Delta\boldsymbol{T}} \Rightarrow \boldsymbol{Q} = \boldsymbol{C}\Delta\boldsymbol{T}$$

Or

$$C = cM$$

- The amount of heat required to change the temperature of **n** moles of a substance, usually for gases, by ΔT is:

$$Q = nC\Delta T$$

Where, n = number of mole
C = Heat capacity

Example 1. In an effort to stay awake for an all night study session a student makes a cup of coffee using an immersion heater in a 0.3kg of water, how much heat must be added to raise the temperature of the water from **20 ºc to 80 ºc**? (take specific heat capacity of water as $\mathbf{c_w}$**=4200J/Kg ºc**) Answer: **Q=75600J**

Latent Heats (Hidden Heat)

- Is the heat required per unit mass of a substance to produce a phase change/change of state at constant temperature.
- The energy required to change the phase of a pure substance of mass m at a constant temperature is given by:

$$Q = \pm mL$$

where **L** is the latent heat of the substance and depends on the nature of the phase change and the properties of the substance. The positive sign is used if energy is entering the system, and the negative sign is used if energy is leaving.

Types of Latent Heat Transfer

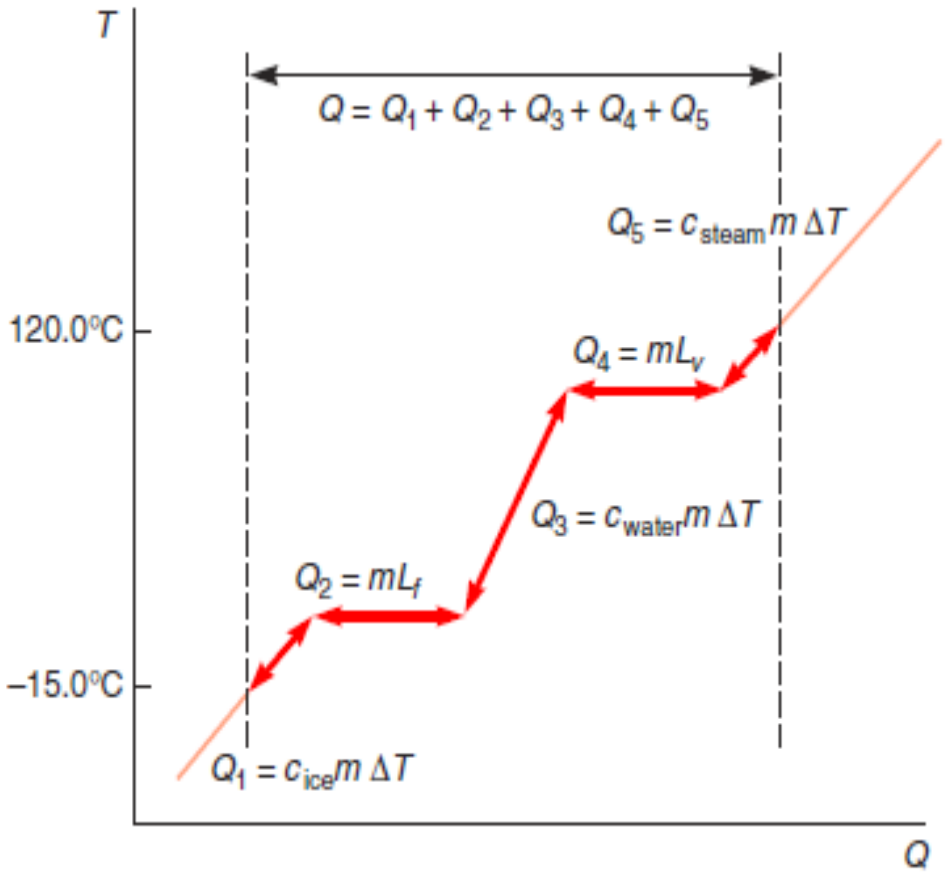
There are two types of latent heat transfers between an object and its environment.

Latent Heat of Fusion (Lf): is the heat absorbed or released when matter melts, changing phase from solid to liquid form at constant temperature. For example, 333.7 kJ of heat is required to change 1 kg of ice to water at 00C, so for water Lf= 333.7kJ/kg.

Latent Heat of Vaporization (L_V): is the heat absorbed or released when matter vaporizes, changing phase from liquid to gas phase at constant temperature. To change 1 kg of water to steam at 100^0C, 2256 kJ of heat is required and so L_V = 2256 kJ.

Example 1. How many joules of heat are needed to change 3.50 kg of ice at -15.0C to steam at 120.0°C?

SOLUTION.



$$Q = c_{\text{ice}} m \Delta T + m L_f + c_{\text{water}} m \Delta T + m L_v + c_{\text{steam}} m \Delta T$$

$$\begin{aligned} Q = {} & \left(2100 \frac{\text{J}}{\cancel{\text{kg}}\,\cancel{^\circ\text{C}}}\right)(3.50\ \cancel{\text{kg}})(15.0\cancel{^\circ\text{C}}) + (3.50\ \cancel{\text{kg}})\left(335 \frac{\text{kJ}}{\cancel{\text{kg}}}\right) \times \frac{10^3\ \text{J}}{1\ \text{kJ}} \text{(Change to joules.)} \\ & + \left(4190 \frac{\text{J}}{\cancel{\text{kg}}\,\cancel{^\circ\text{C}}}\right)(3.50\ \cancel{\text{kg}})(100.0\cancel{^\circ\text{C}}) + (3.50\ \cancel{\text{kg}})\left(2.26 \frac{\text{MJ}}{\cancel{\text{kg}}}\right) \times \frac{10^6\ \text{J}}{1\ \text{MJ}} \\ & + \left(2000 \frac{\text{J}}{\cancel{\text{kg}}\,\cancel{^\circ\text{C}}}\right)(3.50\ \cancel{\text{kg}})(20.0\cancel{^\circ\text{C}}) \\ = {} & 1.080 \times 10^7\ \text{J} \quad \text{or} \quad 10.80\ \text{MJ} \end{aligned}$$

Example 2. How much energy is required to convert a 1.00-g cube of ice at - 30.0°C to steam at 120.0°C ?

Part A. On this portion of the curve, the temperature of the ice changes from − 30.0°C to 0.0°C. Because the specific heat of ice is 2 090 J/kg · °C, we can calculate the amount of energy added by using Equation 20.4:

$$Q = m_i c_i \, \Delta T = (1.00 \times 10^{-3}\ \text{kg})(2\,090\ \text{J/kg} \cdot {}^\circ\text{C})(30.0^\circ\text{C}) = 62.7\ \text{J}$$

Part B. When the temperature of the ice reaches 0.0°C, the ice–water mixture remains at this temperature—even though energy is being added—until all the ice melts. The energy required to melt 1.00 g of ice at 0.0°C is, from Equation 20.6,

$$Q = m_i L_f = (1.00 \times 10^{-3}\ \text{kg})(3.33 \times 10^5\ \text{J/kg}) = 333\ \text{J}$$

Thus, we have moved to the 396 J (= 62.7 J + 333 J) mark on the energy axis in Figure 20.2.

Part C. Between 0.0°C and 100.0°C, nothing surprising happens. No phase change occurs, and so all energy added to the water is used to increase its temperature. The amount of energy necessary to increase the temperature from 0.0°C to 100.0°C is

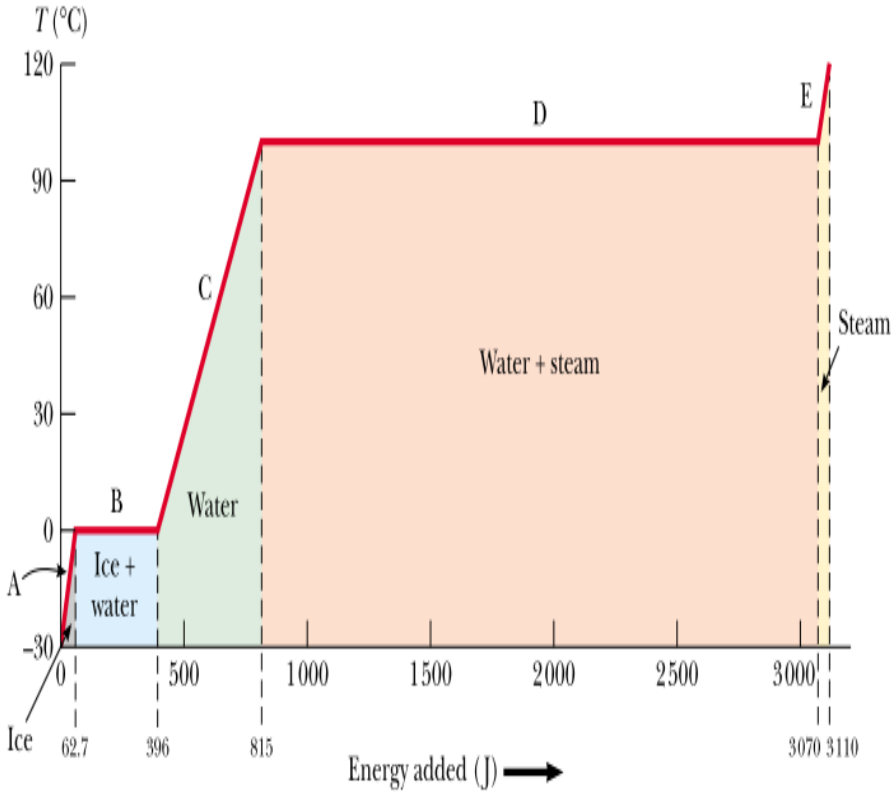
$$Q = m_w c_w \, \Delta T = (1.00 \times 10^{-3}\ \text{kg})(4.19 \times 10^3\ \text{J/kg} \cdot {}^\circ\text{C})(100.0^\circ\text{C}) = 419\ \text{J}$$

Part D. At 100.0°C, another phase change occurs as the water changes from water at 100.0°C to steam at 100.0°C. Similar to the ice–water mixture in part B, the water–steam mixture remains at 100.0°C—even though energy is being added—until all of the liquid has been converted to steam. The energy required to convert 1.00 g of water to steam at 100.0°C is

$$Q = m_w L_v = (1.00 \times 10^{-3}\ \text{kg})(2.26 \times 10^6\ \text{J/kg}) = 2.26 \times 10^3\ \text{J}$$

Part E. On this portion of the curve, as in parts A and C, no phase change occurs; thus, all energy added is used to increase the temperature of the steam. The energy that must be added to raise the temperature of the steam from 100.0°C to 120.0°C is

$$Q = m_s c_s \, \Delta T = (1.00 \times 10^{-3}\ \text{kg})(2.01 \times 10^3\ \text{J/kg} \cdot {}^\circ\text{C})(20.0^\circ\text{C}) = 40.2\ \text{J}$$



Total Heat = Q1 +Q2 +Q3+Q4+Q5 = 3110 J

N:B. To cool 1 g of steam at 120.0°C to ice at & -30.0°C, we must remove 3110 J of energy.

Example 3.

How much heat is needed to change 1.5 kg of ice at −20°C and 1.0 atm into steam?

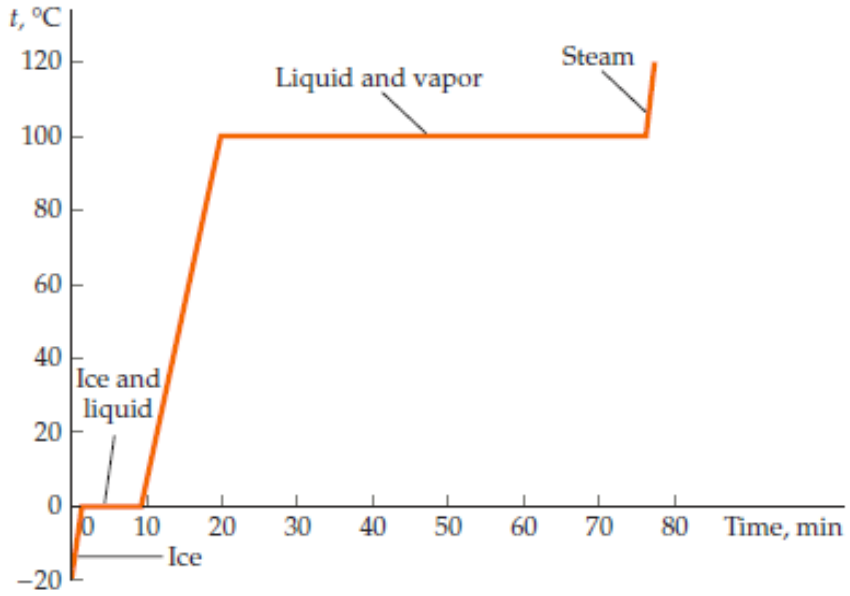


FIGURE 18-1 A 1.5-kg sample of water is heated from −20°C to 120°C at a constant rate of 60 kJ/min.

PICTURE The heat required to change the ice into steam consists of four parts: Q_1, the heat needed to increase the temperature of the ice from −20°C to 0°C; Q_2, the heat needed to melt the ice; Q_3, the heat needed to increase the temperature of the water from 0°C to 100°C; and Q_4, the heat needed to vaporize the water. In calculating Q_1 and Q_3, we assume that the specific heats are constant, with the values 2.05 kJ/kg · K for ice and 4.18 kJ/kg · K for water.

SOLVE

1. Use $Q_1 = mc\,\Delta T$ to find the heat needed to increase the temperature of the ice to 0°C:

$$Q_1 = mc\,\Delta T = (1.5\text{ kg})(2.05\text{ kJ/kg}\cdot\text{K})(20\text{ K}) = 61.5\text{ kJ} = 0.0615\text{ MJ}$$

2. Use L_f from Table 18-2 to find the heat Q_2 needed to melt the ice:

$$Q_2 = mL_f = (1.5\text{ kg})(333.5\text{ kJ/kg}) = 500\text{ kJ} = 0.500\text{ MJ}$$

3. Find the heat Q_3 needed to increase the temperature of the water from 0°C to 100°C:

$$Q_3 = mc\,\Delta T = (1.5\text{ kg})(4.18\text{ kJ/kg}\cdot\text{K})(100\text{ K}) = 627\text{ kJ} = 0.627\text{ MJ}$$

$$Q_3 = 627\text{ kJ} = 0.627\text{ MJ}$$

4. Use L_v from Table 18-2 to find the heat Q_4 needed to vaporize the water:

$$Q_4 = mL_v = (1.5\text{ kg})(2.26\text{ MJ/kg}) = 3.39\text{ MJ}$$

5. Sum your results to find the total heat Q:

$$Q = Q_1 + Q_2 + Q_3 + Q_4 = \boxed{4.6\text{ MJ}}$$

Example 4.

How much heat energy is required to change a 40 g ice cube from a solid at -10 °C to steam at 110 °C?

Solution:

To raise the temperature of the ice to °C we need,

$$\Delta Q_{ice} = m_{ice} c_{ice} \Delta T = 0.04 \text{ kg} \left(0.49 \frac{\text{kcal}}{\text{kg . °C}}\right) 10 \text{ °C} = 0.196 \text{ kcal.}$$

To melt the ice we need,

$$\Delta Q_{ice} = m_{ice} L_{ice} = 0.04 \text{ kg (80 kcal/kg)} = 3.2 \text{ kcal.}$$

To raise the temperature of the water to 100 °C we need,

$$\Delta Q_{water} = m_{water} c_{water} \Delta T = 0.04 \text{ kg} \left(1 \frac{\text{kcal}}{\text{kg °C}}\right) 100 \text{ °C} = 4 \text{ kcal.}$$

To boil the water we need,

$$\Delta Q_L = m_{water} L_{iwater} = 0.04 \text{ kg} \left(540 \text{ k} \frac{\text{cal}}{\text{kg}}\right) = 21.6 \text{ kcal}$$

To raise the temperature of the steam to 110°C we need,

$$\Delta Q_{ice} = m_{steam} c_{steam} \Delta T = 0.04\text{kg}(0.48 \text{ kcal/(kg°C)})10\text{°C} = 0.192 \text{ kcal.}$$

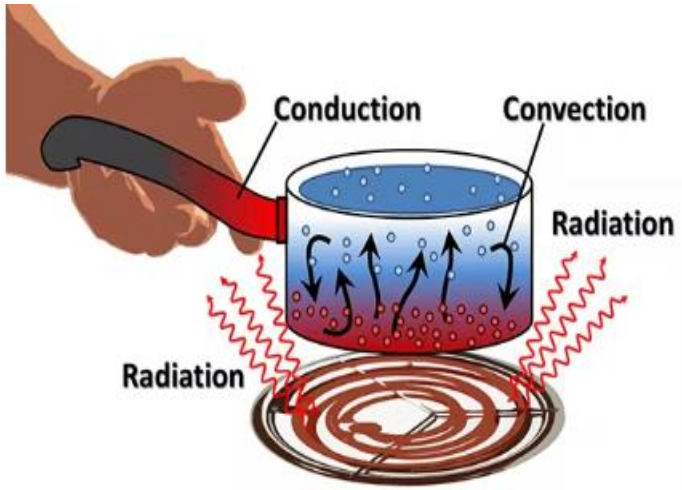
Therefore, the total heat energy required is

$$\Delta Q = (0.196 + 3.2 + 4 + 21.6 + 0.192)\text{kcal} = 29.188 \text{ kcal.}$$

Heat Transfer Mechanisms

Heat may be transferred from one place to another in three ways:

- conduction
- convection
- radiation



1. Conduction :

- transfers heat via direct molecular collision (occurs with physical contact)
- exchange of kinetic energy between colliding molecules or electrons
- is most obvious in solids.
- When a solid heats up, its particles gain kinetic energy and increase the energy with which they vibrate.
- occurs when heat energy travels through a body, passing from particle to particle as they vibrate & collide against each other.

For example,

✓ place your hand against a window or place metal into an open flame.

if you hold one end of a long metal bar and insert the other end into a flame, you will find that the temperature of the metal in your hand soon increases. The energy reaches your hand by means of conduction due to the collision of particles inside the metal.

2. Convection :

- Is the transfer of heat by the movement of the heated particles themselves.
- This can only take place in liquids and gases because in solids the particles are not able to move from their fixed positions.
- When a liquid or gas is heated, it expands and becomes less dense. The lighter liquid or gas rises allowing a flow of cooler material to take its place. This in turn becomes heated and so a current is set up. Heat will continue to be transferred through the available space in this way until it is evenly distributed.

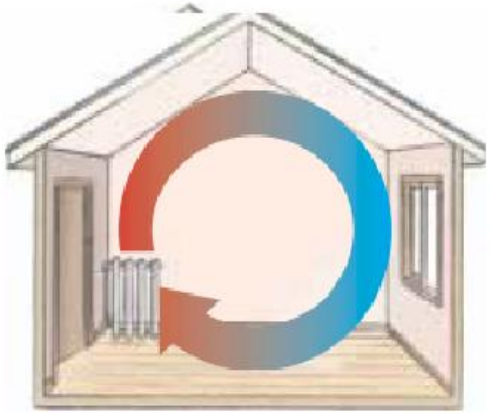


Fig Convection currents

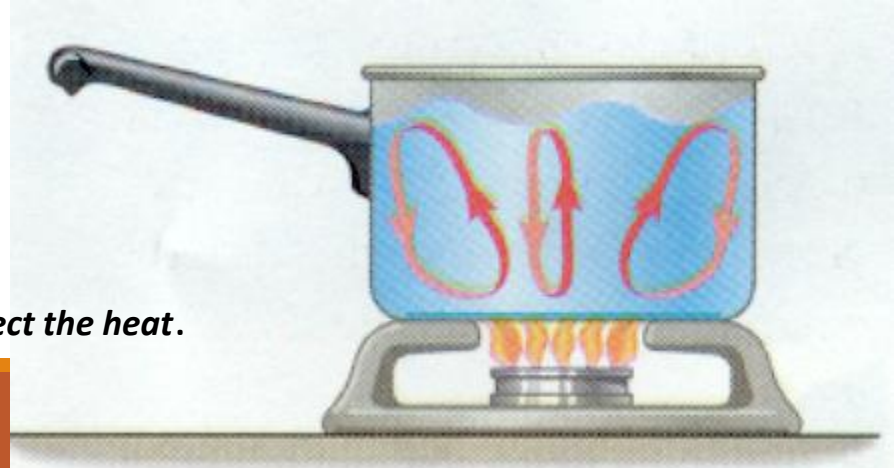
Examples:

✓ Room is heated by a radiator.

- The hot radiator warms the air in the lower regions of the room. The warm air expands and rises to the ceiling because of its lower density. The denser, cooler air from above sinks, and the continuous air current pattern shown in Fig

✓ Water heated in a teakettle

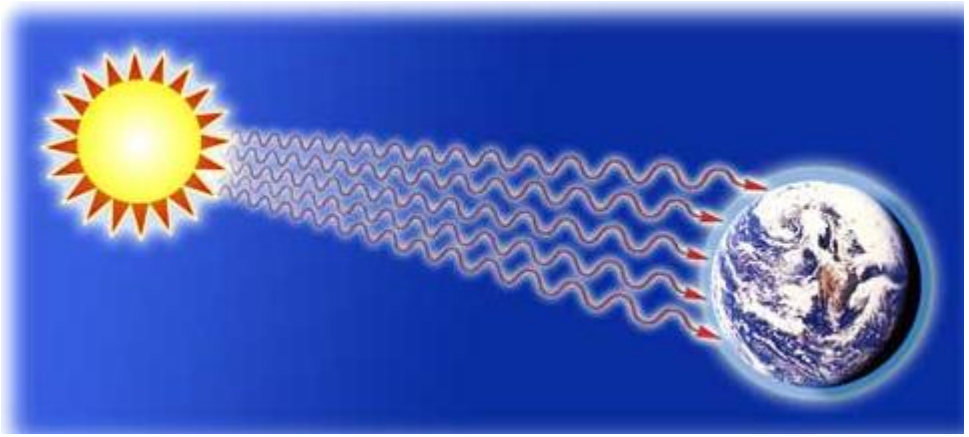
- the lower layers are warmed first. This water expands and rises to the top because its density is lowered. At the same time, the denser, cool water at the surface sinks to the bottom of the kettle and is heated.

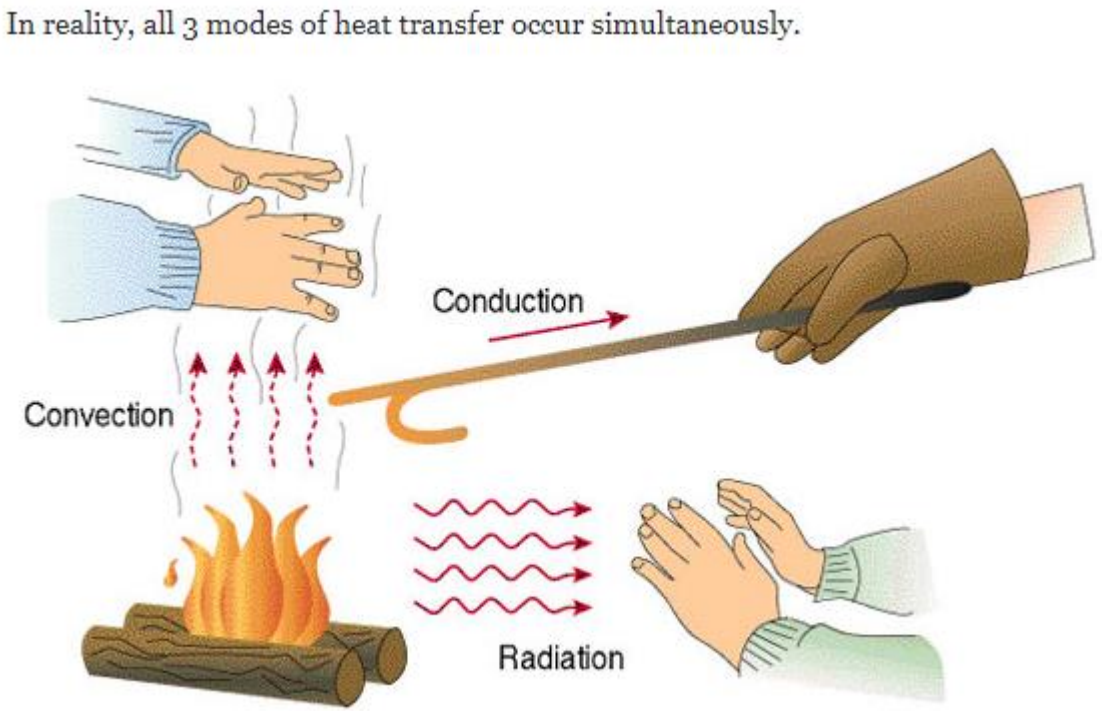


❖ ***the warmer particle moved away from the source of heat to let the cooler ones collect the heat.***

3. Radiation :

- Is the way we receive heat energy from the sun.
- It does not require a medium for its transmission (i.e. it can travel through empty space) and is in the form of electromagnetic energy waves which travel in the same way as light or radio waves





Difference Between Conduction Convection and Radiation

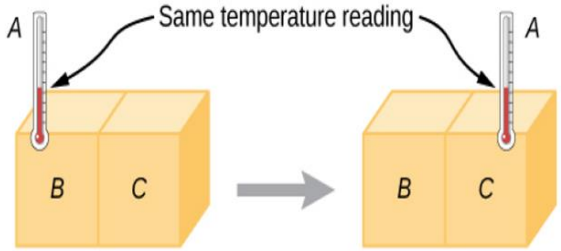
Conduction	Convection	Radiation
In conduction, heat transfer occurs between objects by direct contact.	In convection, the heat transfer takes within the fluid.	In radiation, heat transfer occurs through electromagnetic waves without involving particles.
The heat transfer takes place due to the difference in temperature.	Heat transfer occurs due to the difference in density.	The heat transfer occurs in all objects with a temperature greater than 0 K.
Heat transfer in conduction is slow	Heat transfer in convection is faster.	Heat transfer in radiation is the fastest.
Heat transfer occurs through a heated solid object.	Heat transfer occurs through intermediate objects. For example, heat transfer between air and water.	Heat transfer occurs through electromagnetic waves.
It does not follow the law of reflection and refraction.	It does not follow the law of reflection and refraction.	It follows the law of reflection and refraction.

Law of Thermodynamics

The laws of thermodynamics are extremely powerful tools that will help us to predict the behavior of systems like machines, engines and in particular gases.

1. Zeroth law of Thermodynamics

It states that if two systems are each in thermal equilibrium with a third one, they are also in thermal equilibrium with each other, which means if objects A is in thermal equilibrium with object B and object B is in thermal equilibrium with object C, then object A is in thermal equilibrium with object C.



2. The First Law of Thermodynamics

- states that the change in internal energy of a system is equal to the sum of the heat flow into the system and the work done on the system.

$$\Delta U = Q + W$$

Where ΔU is the change in internal energy of the system, W= PΔV is the work done on the system (or by the system) and P and ΔV are the pressure and change in volume of the system.

- The first law is a specialized statement of energy conservation applied to a thermodynamic system, such as a gas inside a cylinder that has a movable piston.

3. Application of first Law of thermodynamics

Isolated system-is a system which does not exchange heat with its surrounding and no work is done on the external environment. In this case Q=0 and W=0, so from the first law we conclude $\Delta U = 0 \; or \; U = constant$

The internal energy of an isolated system is constant.

Cyclic Process

Engines operate in cycles, in which the system –for example, a gas-periodically returns to its initial state. Since the system returns to its initial state, the change in internal energy in one complete cycle is zero; that is, $\Delta U = 0$. From the first law we see that

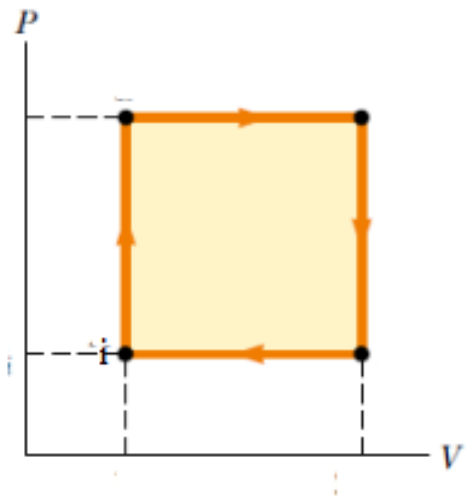
$$Q = -W$$

- In a **cyclic process** (one that originates and terminates at the same state), $\Delta U = 0$ and, therefore, $Q + W = 0 \implies Q = -W$. That is, the energy transferred into the system by heat equals the negative of the work done on the system during the process.

- A **cyclic process** appears as a closed curve/path on a *PV* diagram.

- The net work done on the system per cycle is equal to the negative of the area enclosed by the representing the process

i.e. **area enclosed by the cyclic process= - total work done on the system**



A cyclic process

Isochoric process

In a constant volume process, the volume of the system stays constant. Consequently, W=0. From the first law we see that

$$\Delta U = Q$$

All the heat entering the system goes into increasing the internal energy.

Note: Isovolumetric (Isochoric) Process :

- is one that occurs at constant volume.
- No work is done in such a process (W = 0)

so $\Delta U = Q$

Adiabatic Process

In an adiabatic process, the system does not exchange heat with its surroundings; that is, Q = 0. The first law for an adiabatic process takes the form

$$\Delta U = W$$

- In an **adiabatic process**, no energy is transferred by heat between the system and its surroundings ($Q = 0$). In this case, the first law gives $\Delta U = W$. That is, the internal energy changes as a consequence of work being done on the system. In the adiabatic free expansion of a gas $Q = 0$and $W = 0$, and so $\Delta U = 0$. That is, the internalenergy of the gas does not change in such a process.

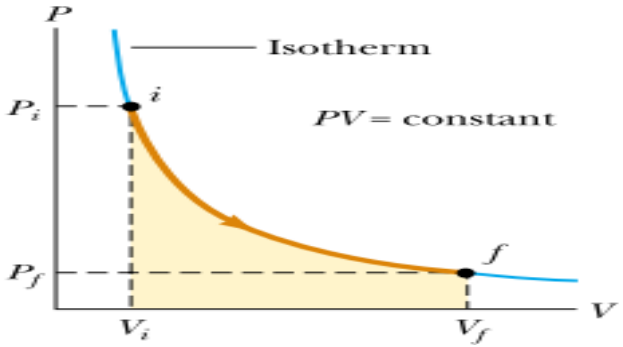
Isothermal Process

It is a process which involves no change in the temperature of the system. If the process occurs at constant temperature then there is no change in the internal energy of the system so ΔU. The first law for an isothermal process takes the form

$$\Delta U = Q + W$$

$$0 = Q + W$$

$$Q = -W$$



For an ideal gas in isothermal process the work done is calculated as

$$W = nRT \ln\left(\frac{V_i}{V_f}\right)$$

R = 8.31 J/mol.K = gas constant

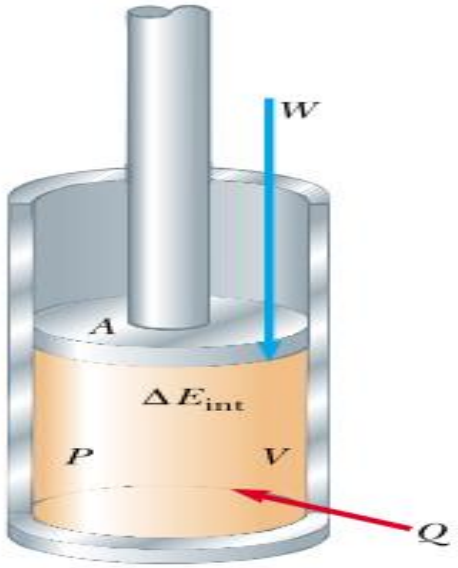
Isobaric process

In an isobaric process the expansion or compression occurs at constant pressure. Any work done by the system will result in an increase in volume. The work done in Pressure- Volume graph is equal to the area under the PV graph. For an isobaric process the work done W is calculated as

$$W = -P\Delta V = -P\,(V_f - V_i)$$

The first law for an isobaric process can be written as

$$\Delta U = Q + W \text{ or } \Delta U = Q - P\Delta V = Q - P\,(V_f - V_i)$$



Example 1

Quick Quiz #1 Characterize the paths in Figure as isobaric, isovolumetric, isothermal, or adiabatic. Note that $Q = 0$ for path B.

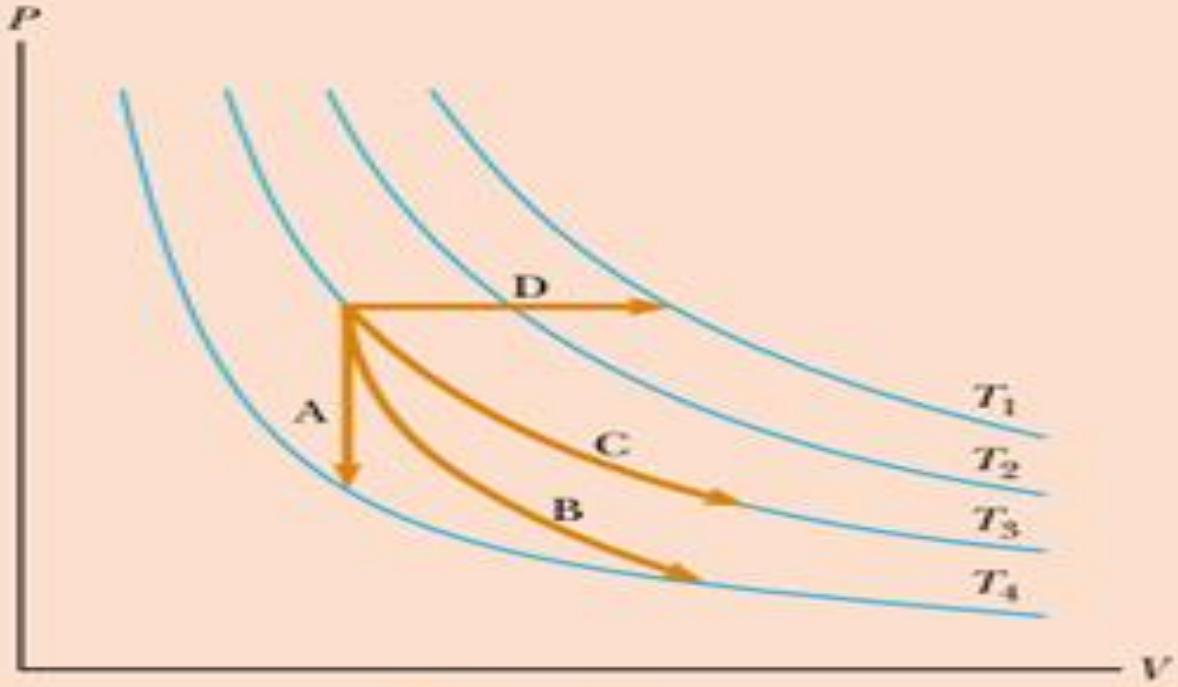


Figure Identify the nature of paths A, B, C, and D.

Answer:

A = Isovolumetric (Isochoric), D = Isobaric, C = Isothermal

B = Adiabatic, b/c Q = 0 implies, $\Delta U = W$ and From this result, we see that if a gas is compressed adiabatically such that W is positive, then ΔU is positive and the temperature of the gas increases. Conversely, the temperature of a gas decreases when the gas expands adiabatically.

Example 2:

2000 J of heat leaves the system and 3000 J of work is done on the system. What is the change in internal energy of the system?

Solution :

$\Delta U = Q + W$

$\Delta U = -2000 + 3000$

$\Delta U = 1000$ Joule

Example3. A 1.0-mol sample of an ideal gas is kept at 0.0°C during an expansion from 3.0 L to 10.0 L

A) How much work is done by the gas during the expansion?

B) How much energy transfer by heat occurs with the surroundings in this process?

C) If the gas is returned to the original volume by means of an isobaric process, how much work is done on the gas?

Solution

Given: V_i = 3L, V_f = 10 L, n =10mol, $T_i = 0^oC$= 273 K = constant(Isothermal), $R = \frac{8.31J}{mol.K}$, $1lit = 1x10^{-3}m^3$

A. $$W = nRT\ln\left(\frac{V_i}{V_f}\right) = (1.0\text{ mol})(8.31\text{ J/mol}\cdot\text{K})(273\text{ K})\ln\left(\frac{3.0\text{ L}}{10.0\text{ L}}\right)$$

$$= -2.7 \times 10^3\text{ J}$$

B. **Solution** From the first law, we find that

$$\Delta E_{int} = Q + W$$
$$0 = Q + W$$
$$Q = -W = 2.7 \times 10^3\text{ J}$$

C. The work done in an isobaric process is given .

In this case, the initial volume is 10.0 L and the final volume is 3.0 L, the reverse of the situation in part (A). We are not given the pressure, so we need to incorporate the ideal gas law:

$$W = -P(V_f - V_i) = -\frac{nRT_i}{V_i}(V_f - V_i)$$

$$= -\frac{(1.0\text{ mol})(8.31\text{ J/mol}\cdot\text{K})(273\text{ K})}{10.0 \times 10^{-3}\text{ m}^3} \times (3.0 \times 10^{-3}\text{ m}^3 - 10.0 \times 10^{-3}\text{ m}^3)$$

$$= 1.6 \times 10^3\text{ J}$$

Notice that we use the initial temperature and volume to determine the value of the constant pressure because we do not know the final temperature. The work done on the gas is positive because the gas is being compressed.

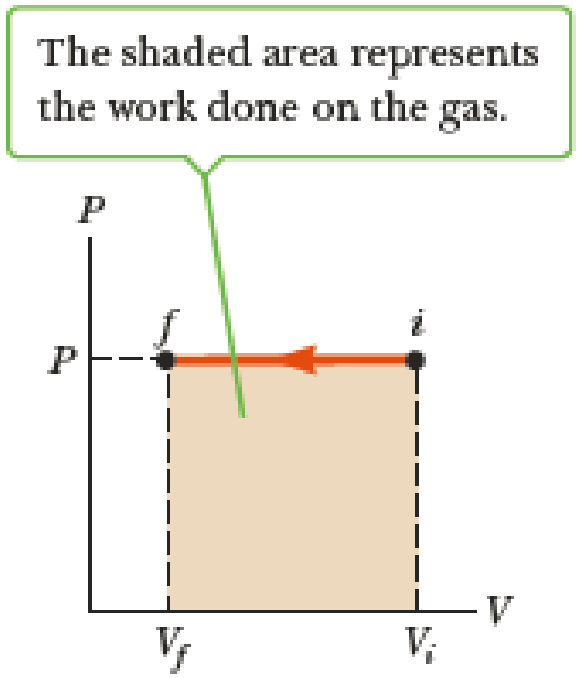
Work done in thermodynamic processes

- The area under the graph in a *PV* diagram is equal to the negative of the work done on the system/gas.

i.e. Area under PV graph = -work done on the system

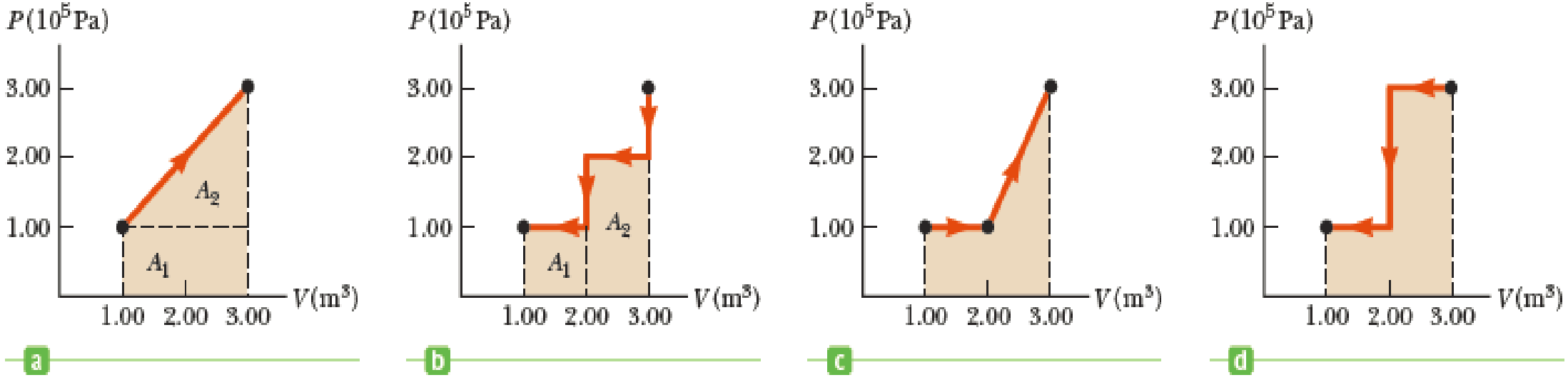
For instance: The *PV* diagram for a gas being compressed at constant pressure is given below. In This case the area under the graph is

$$\text{Area} = P(V_f - V_i) = P\Delta V$$



Examples

1. Find the numeric value of the work done on the gas as it moves from an initial state to a final state as shown in figures **(a)** , **(b) ,(c) and (d)**



solution

(a) Find the work done on the gas in Figure 12.4a.

Compute the areas A_1 and A_2 in Figure 12.4a. A_1 is a rectangle and A_2 is a triangle.

$$A_1 = \text{height} \times \text{width} = (1.00 \times 10^5\ \text{Pa})(2.00\ \text{m}^3)$$
$$= 2.00 \times 10^5\ \text{J}$$
$$A_2 = \tfrac{1}{2}\ \text{base} \times \text{height}$$
$$= \tfrac{1}{2}(2.00\ \text{m}^3)(2.00 \times 10^5\ \text{Pa}) = 2.00 \times 10^5\ \text{J}$$

Sum the areas (the arrows point to increasing volume, so the work done on the gas is negative):

$$\text{Area} = A_1 + A_2 = 4.00 \times 10^5\ \text{J}$$
$$W = -4.00 \times 10^5\ \text{J}$$

(b) Find the work done on the gas in Figure 12.4b.

Compute the areas of the two rectangular regions:

$$A_1 = \text{height} \times \text{width} = (1.00 \times 10^5\ \text{Pa})(1.00\ \text{m}^3)$$
$$= 1.00 \times 10^5\ \text{J}$$
$$A_2 = \text{height} \times \text{width} = (2.00 \times 10^5\ \text{Pa})(1.00\ \text{m}^3)$$
$$= 2.00 \times 10^5\ \text{J}$$

Sum the areas (the arrows point to decreasing volume, so the work done on the gas is positive):

$$\text{Area} = A_1 + A_2 = 3.00 \times 10^5\ \text{J}$$
$$W = +3.00 \times 10^5\ \text{J}$$

Answers for (C) and (d)

$-3.00 \times 10^5\ \text{J}$, $+4.00 \times 10^5\ \text{J}$

2. A gas is taken through the cyclic process described in Figure below.

(a) Find the net energy transferred to the system by heat during one complete cycle.

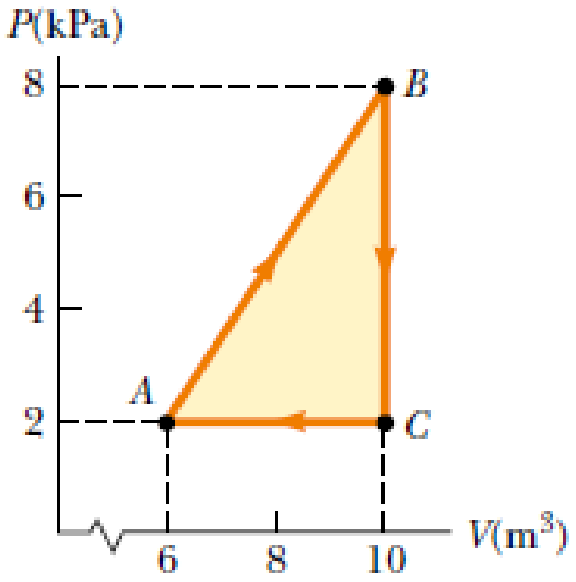
(b) **What If?** If the cycle is reversed—that is, the process follows the path *ACBA*—what is the net energy input per cycle by heat?

solution

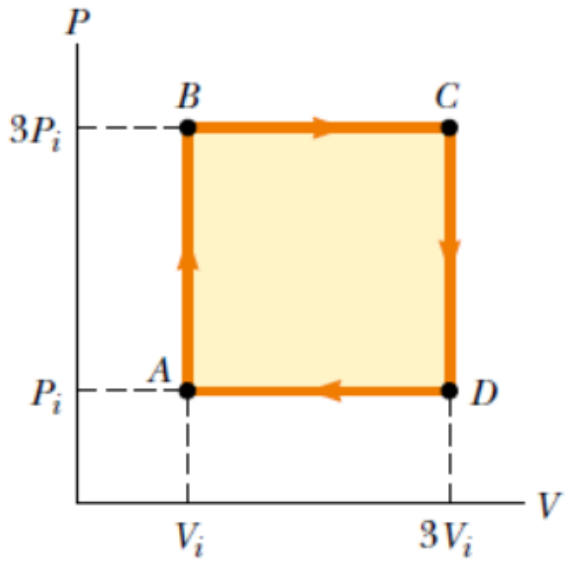
(a) $Q = -W =$ Area of triangle

$$Q = \frac{1}{2}\left(4.00\ \text{m}^3\right)(6.00\ \text{kPa}) = \boxed{12.0\ \text{kJ}}$$

(b) $Q = -W = \boxed{-12.0\ \text{kJ}}$



3. An ideal gas initially at P_i , V_i , and T_i is taken through a cycle as shown in Figure below.
(a) Find the net work done on the gas per cycle.
(b) What is the net energy added by heat to the system per cycle?



solution

(a) The work done during each step of the cycle equals the negative of the area under that segment of the PV curve.

$$W = W_{DA} + W_{AB} + W_{BC} + W_{CD}$$

$$W = -P_i(V_i - 3V_i) + 0 - 3P_i(3V_i - V_i) + 0 = \boxed{-4P_iV_i}$$

(b) The initial and final values of T for the system are equal. Therefore, $\Delta E_{int} = 0$ and $Q = -W = \boxed{4P_iV_i}$.